

William Li

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Education

Simon Fraser University

Bachelor of Science in Software Systems

GPA: 3.2

Sept 2024 – Aug 2028

- Courses: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Compilers

Experience

Software Engineer Intern

Vici Learning

Jun 2026 – Aug 2026

Hybrid

- Collaborated in a 5-person Scrum team to build an internal dashboard for Vici Learning, consolidating **SimplyBook.me**, **Brevo**, and **Notion** into one centralized view, eliminating the need to check three separate systems for different statuses.
- Configured **Spring Security** with **ADMIN** and **TUTOR** roles to protect access to sensitive student and family data.
- Built a hand-written **JSON-RPC 2.0** client on Spring's `RestClient` to sync bookings, clients, services, and tutors from **SimplyBook.me** into a local **PostgreSQL** database via scheduled background jobs, insulating the system from API rate limits and upstream downtime.
- Integrated **Brevo's** REST API to fetch and reconcile contact and family-account data, surfacing live active, paused, and dropped student counts on the dashboard.
- Built the metrics engine powering the staff dashboard, cutting weekly manual cross-referencing time by **54%**

Campaign Technology & Communications Consultant

SD43 School Trustee Election

Aug 2026 – Oct 2026

Hybrid

- Engineered a full-stack campaign website using **React 19**, **TypeScript**, and **Vite**, with serverless **Vercel** functions integrating **Resend** for automated email handling, driving **300+** unique visitors tracked via **Vercel Analytics**.
- Conducted geographic and demographic voter data analysis to identify priority outreach zones, cutting canvassing time by over **11** hours and increasing voter contact by **60%**
- Presented technology roadmaps and delivery plans to the candidate and PR team using agile sprint cycles and velocity tracking, aligning stakeholders across the campaign timeline.

Projects

Escape From Windows – Retro OS-simulated puzzle game | *Java, Python, TypeScript, Node.js, Express, Docker, REST API*

- 🏆 **SystemHacks 2026: Most Outstanding Project**
- Built a full-stack browser-based puzzle game using **React**, **TypeScript**, and a **Node.js/Express REST API** to manage persistent game state, authentication, and session data.
- Developed a **Python-based AI hint engine** that analyzes player interaction patterns and dynamically generates contextual hints using heuristic modeling.
- Containerized the application with **Docker** and **Docker Compose**, ensuring reproducible development and production environments during a 24-hour hackathon.
- Implemented automated CI/CD pipelines with **GitHub Actions**; deployed frontend to **Vercel** and backend services to **Render**.
- Designed a simulated OS-style UI with modular **React component architecture**, enabling real-time puzzle triggers via structured **client-server API communication**.

OnDeck – Office Hours Queue & Analytics Platform | *React, Node.js, Express, PostgreSQL, WebSocket, GCP*

- Architected and built a real-time office hours queuing platform for high-demand university courses, replacing ad hoc Google Docs and Zoom waiting rooms with a self-serve queue.
- Owned authentication and authorization, implementing session-based auth and **Role-Based Access Control (RBAC)** express middleware to separate Student, TA, and Instructor views and permissions.
- Built secure session management and access-control logic supporting real-time **WebSocket** state shared across the web client.
- Implemented a **Node.js/Express REST API** on **PostgreSQL**, deployed to a Google Cloud VM behind an **Nginx** reverse proxy, and integrated with the **Google Calendar API** for session scheduling and notifications.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C, C++, SQL, Bash, HTML, CSS

Frameworks: React, Next.js, Express, Node.js, React Native, REST API, FastAPI, Tailwind

Technologies: Git, Docker, GCP, Linux, PostgreSQL, MySQL, Nginx, MongoDB, SQLite